

DAY 3: ORDINARY DIFFERENTIAL EQUATIONS - PROBLEMS

Instructions

Solve all 10 problems showing complete work. All questions are based on Newtonian mechanics and require solving differential equations. Include units in your answers. Box your final answers.

Problem 1: Free Fall with Linear Air Resistance

A skydiver of mass $m = 80$ kg jumps from rest. The air resistance is proportional to velocity: $F_{\text{air}} = -kv$ where $k = 15$ N·s/m. The equation of motion is:

$$m \frac{dv}{dt} = mg - kv$$

- This is a first-order linear ODE. Solve it for $v(t)$ with initial condition $v(0) = 0$.
- Find the terminal velocity v_{term} .
- How long does it take to reach 95% of terminal velocity?
- Integrate $v(t)$ to find the position $y(t)$ if $y(0) = 0$.

Problem 2: Free Fall with Quadratic Air Resistance

For high speeds, air resistance is often proportional to the square of velocity. For a falling object:

$$m \frac{dv}{dt} = mg - kv^2$$

where $k = 0.25$ kg/m, $m = 5$ kg, and $g = 9.8$ m/s². Assume the object starts from rest.

- This is a separable ODE. Separate variables and set up the integral.
- Use the integral formula $\int \frac{dv}{a^2 - v^2} = \frac{1}{2a} \ln \left| \frac{a+v}{a-v} \right| + C$ (where $a = \sqrt{mg/k}$) to solve for $v(t)$.
- Find the terminal velocity.
- Find the time to reach 99% of terminal velocity.

Problem 3: Simple Harmonic Oscillator

A mass $m = 0.5$ kg is attached to a spring with spring constant $k = 8$ N/m. The mass is displaced 0.2 m from equilibrium and released from rest. The equation is:

$$m \frac{d^2 x}{dt^2} = -kx$$

- Write this as $\frac{d^2 x}{dt^2} + \omega^2 x = 0$ and find ω .
- Solve the ODE with initial conditions $x(0) = 0.2$ m, $v(0) = 0$.
- Find the period of oscillation.
- Find the maximum velocity and maximum acceleration.

Problem 4: Vertical Spring

A mass $m = 2$ kg hangs from a vertical spring with spring constant $k = 50$ N/m. Let y be measured downward from the unstretched position.

- Write the equation of motion including gravity.
- Find the equilibrium position y_0 where $mg = ky_0$.
- Define a new coordinate $z = y - y_0$. Show that z satisfies the simple harmonic oscillator equation.
- Solve for $z(t)$ if the mass is pulled down 0.1 m below equilibrium and released from rest.
- Find the period and frequency of oscillation.

Problem 5: Damped Harmonic Oscillator - Underdamped

A mass $m = 1$ kg is attached to a spring with $k = 100$ N/m and a damper with $b = 2$ N·s/m. The equation is:

$$m \frac{d^2 x}{dt^2} + b \frac{dx}{dt} + kx = 0$$

- Find the damping coefficient $\beta = b/(2m)$ and the natural frequency $\omega_0 = \sqrt{k/m}$.
- Determine if the system is underdamped, critically damped, or overdamped.
- Find the damped frequency $\omega_1 = \sqrt{\omega_0^2 - \beta^2}$.
- Solve for $x(t)$ if $x(0) = 0.1$ m and $v(0) = 0$.
- Find the first three times when the mass passes through $x = 0$.

Problem 6: Damped Harmonic Oscillator - Overdamped

A mass $m = 1$ kg is attached to a spring with $k = 10$ N/m and a damper with $b = 8$ N·s/m.

- Find β , ω_0 , and determine the damping regime.
- Find the two real roots r_1 and r_2 of the characteristic equation.
- Solve for $x(t)$ if $x(0) = 0.1$ m and $v(0) = 0$.
- Solve for $x(t)$ if $x(0) = 0.1$ m and $v(0) = -0.5$ m/s (pushed downward).
- Compare the behavior in the two cases.

Problem 7: Damped Harmonic Oscillator - Critically Damped

For the same mass $m = 1$ kg and spring $k = 16$ N/m, find the damping constant b that makes the system critically damped.

- What is the critical damping value b_c ?
- Write the general solution for $x(t)$ in the critically damped case.
- Solve for $x(t)$ if $x(0) = 0.1$ m and $v(0) = 0$.

- d. Solve for $x(t)$ if $x(0) = 0.1$ m and $v(0) = -0.5$ m/s.
- e. Show that the critically damped system returns to equilibrium faster than the overdamped system for the same initial conditions. (Compare the times to reach $x = 0.01$ m)

Problem 8: Driven Harmonic Oscillator (Resonance)

An undamped harmonic oscillator with $m = 1$ kg, $k = 100$ N/m is driven by an external force $F(t) = F_0 \cos(\omega t)$ with $F_0 = 5$ N.

- a. Find the natural frequency ω_0 .
- b. For a driving frequency $\omega = 8$ rad/s (below resonance), find the particular solution $x_p(t)$.
- c. For a driving frequency $\omega = 12$ rad/s (above resonance), find $x_p(t)$.
- d. For $\omega = \omega_0$ (resonance), find the particular solution. What happens to the amplitude?
- e. Explain physically why the amplitude grows at resonance.

Problem 9: Damped Driven Oscillator

A damped oscillator with $m = 2$ kg, $k = 200$ N/m, $b = 4$ N·s/m is driven by $F(t) = 10 \cos(\omega t)$ N.

- a. Find β , ω_0 , and the damping ratio $\zeta = \beta/\omega_0$.
- b. Find the particular solution for $\omega = 8$ rad/s. (Try $x_p = A \cos(\omega t) + B \sin(\omega t)$)
- c. Find the amplitude $C = \sqrt{A^2 + B^2}$ as a function of ω .
- d. At what driving frequency ω is the amplitude maximized? (This is the resonance frequency for a damped system)
- e. Find the maximum amplitude.